Anelench: mathematics, algorithms and physics

The derivations, proofs and procedures behind the counterexample search and the observation planner

Companion to the plain-language explanation. Every formula here is the one the code evaluates; where the code makes an approximation, the approximation is stated.

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1. Notation and conventions

Sky offsets are measured from the image centre in microarcseconds (μas): x increases towards the east and y towards the north, so a sky image is drawn with east on the left. The image is a grid of $N_{\mathrm{pix}} = 64 \times 64$ pixels covering a field of view of $128\,\mu\mathrm{as}$ (pixel size $\Delta = 2\,\mu\mathrm{as}$). Pixel (r, c) sits at

$$x_c = \left(\tfrac{n}{2} - \tfrac{1}{2} - c\right)\Delta, \qquad y_r = \left(\tfrac{n}{2} - \tfrac{1}{2} - r\right)\Delta, \qquad n = 64.$$

Position angles (PA) are measured from north through east. Spatial frequencies (u, v) are in wavelengths ($1\,\mathrm{G}\lambda = 10^9$ wavelengths). $1\,\mu\mathrm{as} = 4.8481 \times 10^{-12}\,\mathrm{rad}$. An image I has total flux $F = \sum_j I_j$; we fix $F = 0.6$ Jy, the compact flux of M87* used by the EHT. All closure quantities are invariant to F, so this is a normalisation, not an assumption.

2. The astrophysical setting

2.1 Angular size of the black hole

For a black hole of mass M at distance D, the natural angular unit is the gravitational radius seen from Earth,

$$\theta_g = \frac{GM}{c^2 D}.$$

With $M = 6.5 \times 10^9\, M_\odot$ and $D = 16.8$ Mpc: $GM/c^2 = 6.5 \times 10^9 \times 1.4766\,\mathrm{km} = 9.60 \times 10^{12}$ m and $D = 5.18 \times 10^{23}$ m, so $\theta_g = 1.85 \times 10^{-11}$ rad $= 3.8\,\mu$as.

2.2 The photon ring

For a non-rotating (Schwarzschild) black hole, a photon with energy E and angular momentum L moving in the equatorial plane obeys, in units $G = c = 1$,

$$\left(\frac{dr}{d\lambda}\right)^2 = E^2 - \frac{L^2}{r^2}\left(1 - \frac{2M}{r}\right).$$

The right-hand side stays positive at all radii only if $1/b^2$ exceeds the maximum of $V(r) = r^{-2}(1 - 2M/r)$, where $b = L/E$ is the impact parameter. Setting $V'(r) = 0$ gives $r = 3M$ (the photon sphere) and $V(3M) = 1/(27M^2)$. Photons with $b < b_c = \sqrt{27}\,M$ are captured, so the dark shadow has diameter

$$2b_c\,\frac{1}{D} = 2\sqrt{27}\,\theta_g \approx 10.4\,\theta_g \approx 39.7\,\mu\mathrm{as}.$$

Spin changes this by only a few per cent; the EHT measured a ring of $42 \pm 3\,\mu$as. This is why our synthetic sources are rings of 38–46 μas diameter.

2.3 Resolution of an Earth-sized array

At $\nu = 228$ GHz ($\lambda = c/\nu = 1.31$ mm) the longest 2017 baseline reaches $|\mathbf{u}|_{\max} \approx 8.3\,\mathrm{G}\lambda$, so the finest fringe spacing is $1/|\mathbf{u}|_{\max} \approx 25\,\mu$as. Structure smaller than this is not measured directly; it is inferred under assumptions. A knot 40 μas from the ring centre is therefore only about one resolution element outside the ring: exactly the regime where images can disagree.

2.4 Timescales

The gravitational time $t_g = GM/c^3 = 9.60 \times 10^{12}\,\mathrm{m}/c = 3.2 \times 10^4$ s ≈ 8.9 h. One observing night is about one t_g, so a static image is a fair model within a night; six days (April 5 to April 11) are about 16 t_g, so structure may change between nights. Anelench therefore treats other nights as follow-ups of a static source only as an assumption, and says so.

2.5 Why a knot near the jet base

M87 launches a relativistic jet at position angle about 288°. Emission just outside the ring along that direction would be evidence about where and how the jet is loaded. A claim of that kind is a statement about faint structure one beam outside the ring; it is the kind of claim Anelench is built to test.

3. What an interferometer measures

3.1 The visibility

By the van Cittert–Zernike theorem, a pair of telescopes measures one Fourier component of the sky brightness:

$$V(u, v) = \iint I(x, y)\, e^{-2\pi i (ux + vy)}\, dx\, dy.$$

The full expression contains a term in w proportional to $1-\sqrt{1-x^2-y^2}\approx (x^2+y^2)/2$. Over our field, $x^2+y^2\le (6.2\times10^{-10})^2$, so with $|w|<10^{10}$ the extra phase is below 2×10^{-9} turns: negligible.

On the pixel grid the integral becomes a discrete Fourier transform, evaluated exactly (no gridding) because there are only a few hundred (u,v) points:

$$V_k=\sum_{j=1}^{N_{\rm pix}} I_j\, e^{-2\pi i(u_k x_j+v_k y_j)}=\sum_j A_{kj} I_j .$$

Because I is real, $V(-u,-v)=\overline{V(u,v)}$: every measurement also fixes the point opposite the origin, which is why coverage plots show symmetric tracks.

3.2 Baseline coordinates from Earth rotation

Let $\mathbf{b}=\mathbf{r}_1-\mathbf{r}_2$ be the baseline between two stations in an Earth-fixed frame (X towards longitude 0 on the equator, Y towards 90° E, Z towards the north pole), δ the source declination and H its Greenwich hour angle, $H=\mathrm{GMST}-\alpha$. The source direction is $\hat{\mathbf{s}}=(\cos\delta\cos H,\ -\cos\delta\sin H,\ \sin\delta)$. The projected coordinates use the orthonormal basis

$$\hat{\mathbf{u}}=(\sin H,\ \cos H,\ 0),\quad \hat{\mathbf{v}}=(-\sin\delta\cos H,\ \sin\delta\sin H,\ \cos\delta),\quad \hat{\mathbf{w}}=\hat{\mathbf{s}},$$

so that, in wavelengths,

$$u=\frac{\mathbf{b}\cdot\hat{\mathbf{u}}}{\lambda}=\frac{b_X\sin H+b_Y\cos H}{\lambda},\qquad v=\frac{\mathbf{b}\cdot\hat{\mathbf{v}}}{\lambda}=\frac{-b_X\sin\delta\cos H+b_Y\sin\delta\sin H+b_Z\cos\delta}{\lambda}.$$

One checks $\hat{\mathbf{u}}\cdot\hat{\mathbf{s}}=\hat{\mathbf{v}}\cdot\hat{\mathbf{s}}=0$, $|\hat{\mathbf{u}}|=|\hat{\mathbf{v}}|=1$. The Greenwich mean sidereal time is

$$\mathrm{GMST}=280.46061837^\circ+360.98564736629^\circ\,(J-2451545)+0.000387933\,T^2-\frac{T^3}{38710000},$$

with J the Julian date and $T=(J-2451545)/36525$. The source elevation at a station with geodetic latitude ϕ and east longitude ℓ follows from the spherical cosine rule with local hour angle $h=\mathrm{GMST}+\ell-\alpha$:

$$\sin e=\sin\phi\sin\delta+\cos\phi\cos\delta\cos h.$$

A station observes when $e\ge10^\circ$. The point on Earth where the source is overhead (the sub-source point) has latitude δ and longitude $\alpha-\mathrm{GMST}$; every station on the hemisphere facing it has the source above the horizon. This is the globe in the interface.

The station coordinates are converted between geodetic and Earth-fixed form on the WGS84 ellipsoid (semi-major axis $a=6378137$ m, flattening $f=1/298.257223563$), inverting with Bowring's fixed-point iteration.

3.3 Conventions of the released data

The (u, v) values in the EHT release match $\mathbf{b}=\mathbf{r}_1-\mathbf{r}_2$ above to a relative error of 3×10^{-5} on every baseline. The released visibilities follow the complex-conjugate sign convention of the forward model in 3.1; we conjugate them on loading. The check is physical: with the conjugate, every reconstruction of April 11 has its bright side in the south, as in the published images; without it, in the north.

4. Station gains and closure quantities

Each station adds an unknown complex gain g_i (atmospheric phase, pointing, amplitude calibration), so the measurement on baseline ij is

$$y_{ij} = g_i \, \overline{g_j} \, V_{ij} + \epsilon_{ij}.$$

At 1.3 mm the atmospheric phase changes within seconds, so the phases of y_{ij} are useless on their own. Anelench uses only combinations from which the gains cancel.

Theorem 1 (closure phase). For stations a, b, c and noise-free data, $\psi_{abc} = \arg\left(y_{ab}\, y_{bc}\, y_{ca}\right)$ does not depend on the gains.

Proof. $y_{ab}y_{bc}y_{ca} = g_a\overline{g_b}\, g_b\overline{g_c}\, g_c\overline{g_a}\, V_{ab}V_{bc}V_{ca} = |g_a|^2|g_b|^2|g_c|^2\, V_{ab}V_{bc}V_{ca}$. The prefactor is real and positive, so the argument equals $\arg(V_{ab}V_{bc}V_{ca})$. □

Theorem 2 (log closure amplitude). For four stations, $\ell_{abcd} = \log|y_{ab}| + \log|y_{cd}| - \log|y_{ac}| - \log|y_{bd}|$ does not depend on the gains.

Proof. $\log|y_{ij}| = \log|V_{ij}| + \log|g_i| + \log|g_j|$. Each station index appears once with $+$ and once with $-$ in ℓ_{abcd} (a: ab, ac; b: ab, bd; c: cd, ac; d: cd, bd), so all gain terms cancel. □

4.1 How many independent closures exist?

Write the phases of a scan's baselines as a vector $\boldsymbol{\varphi} \in \mathbb{R}^{|E|}$, where E is the set of baselines with data among S stations. Station phase errors $\boldsymbol{\theta} \in \mathbb{R}^{S}$ enter as $\boldsymbol{\varphi}^{\mathrm{obs}} = \boldsymbol{\varphi} + B\boldsymbol{\theta}$, with the signed incidence matrix $B_{e,i} = +1$ if i is the first station of baseline e, -1 if the second, 0 otherwise. A linear combination $\mathbf{c}^{\mathsf{T}}\boldsymbol{\varphi}$ is gain-invariant exactly when $\mathbf{c}^{\mathsf{T}}B = 0$.

Theorem 3. If the baseline graph is connected, the gain-invariant phase combinations form a space of dimension $|E| - S + 1$. For a complete graph this is $(S-1)(S-2)/2$.

Proof. The dimension of $\{\mathbf{c} : B^{\mathsf{T}}\mathbf{c} = 0\}$ is $|E| - \mathrm{rank}(B)$. If $B\boldsymbol{\theta} = 0$ then $\theta_i = \theta_j$ across every baseline; in a connected graph this forces $\boldsymbol{\theta}$ to be constant, so $\ker B$ is one-dimensional and $\mathrm{rank}(B) = S - 1$. For the complete graph, $S(S-1)/2 - S + 1 = (S-1)(S-2)/2$. □

Amplitude gains enter through the unsigned incidence matrix $M_{e,i} = 1$ if $i \in e$: $\log|\mathbf{y}^{\mathrm{obs}}| = \log|\mathbf{y}| + M\mathbf{a}$.

Theorem 4. If the baseline graph is connected and not bipartite, the gain-invariant log-amplitude combinations form a space of dimension $|E| - S$; for a complete graph with $S \geq 3$, $S(S-3)/2$.

Proof. If $M\mathbf{a} = 0$ then $a_i = -a_j$ across every baseline. Going around a cycle of odd length returns to the start after an odd number of sign changes, giving $a_i = -a_i$, hence $a_i = 0$ on the cycle and, by connectedness, everywhere. So $\mathrm{rank}(M) = S$ and the dimension is $|E| - S$. A complete graph on three or more stations contains a triangle, an odd cycle. □

The same argument proves that per-station sensitivities can be recovered from baseline noise levels (Section 5.2): that problem is M acting on $\log$SEFDs.

4.2 Choosing a basis: greedy is optimal

Within each scan and frequency band, all triangles (and all pairs of matchings of all quadrangles) whose visibilities pass a signal-to-noise cut are candidates. Each candidate is a vector in $\mathbb{R}^{|E|}$ (entries ± 1 on its baselines). Anelench sorts candidates by their noise σ and keeps a candidate whenever its Gram–Schmidt residual against the kept ones is non-zero.

> **Proposition 5.** The kept set is a basis of the span of all candidates, and among all such bases it minimises the total noise $\sum \sigma$.
>
> **Proof.** Linearly independent subsets of a finite set of vectors form a matroid (the linear matroid). For a matroid, the greedy algorithm that scans elements in increasing weight and adds an element whenever independence is preserved returns a minimum-weight basis (Rado–Edmonds theorem). The Gram–Schmidt residual test is exactly the independence test. □

No information is counted twice: the chosen closures are linearly independent functions of the gain-free information in the scan. (Their noises are still correlated through shared baselines; like the standard EHT pipelines we neglect these covariances.)

5. Noise

5.1 Radiometer equation

The thermal noise of one polarisation on baseline ij, per real and imaginary component, is

$$\sigma_{RR} = \frac{1}{\eta_Q}\sqrt{\frac{\mathrm{SEFD}_i\,\mathrm{SEFD}_j}{2\,\Delta\nu\,\tau}},$$

with $\eta_Q = 0.88$ for 2-bit sampling, bandwidth $\Delta\nu$ and integration time τ. Stokes I averages two independent polarisations, so $\sigma_I = \sigma_{RR}/\sqrt{2}$:

$$\sigma_I^2 = \frac{\mathrm{SEFD}_i\,\mathrm{SEFD}_j}{4\,\eta_Q^2\,\Delta\nu\,\tau}.$$

The release sets the errors of RR and LL to $\sqrt{2}$ times the Stokes I error; we undo that factor on reading.

5.2 Effective sensitivities from the release

Taking logarithms of the radiometer equation gives a linear system in the unknown $\log \mathrm{SEFD}$:

$$\log\left(4\eta_Q^2\Delta\nu\,\tau\,\sigma_I^2\right)_{ij} = \log \mathrm{SEFD}_i + \log \mathrm{SEFD}_j,$$

which is M from Theorem 4 acting on the vector of $\log \mathrm{SEFD}$. We use the median over each baseline's 10 s records and solve by least squares; Theorem 4 guarantees a unique solution because the 2017 baseline graph contains triangles. The effective values (which include opacity, elevation and phasing losses) are AA 98, AP 8028, AZ 10771, JC 29250, LM 5028, PV 9701, SM 13834 Jy. For stations added later (GLT, Kitt Peak, NOEMA) nominal values are scaled by the median ratio of effective to nominal SEFD of the 2017 sites.

5.3 Averaging and the systematic budget

Records are averaged coherently within each scan with inverse-variance weights, $\bar V = \sum w_k V_k / \sum w_k$, $w_k = \sigma_k^{-2}$, $\bar\sigma = (\sum w_k)^{-1/2}$. Intra-site baselines (ALMA–APEX, JCMT–SMA) see extended jet emission and are dropped. A fractional non-closing error $f = 2\%$ is added in quadrature, $\sigma^2 \leftarrow \sigma^2 + (f|V|)^2$, as in the EHT analyses. Synthetic data are generated with exactly this budget, so the true image has $\chi^2_\nu \approx 1$.

5.4 Errors of closure quantities

For $y = V + \epsilon$ with complex Gaussian ϵ of standard deviation σ per component, to first order $\delta \arg y = \mathrm{Im}(\epsilon/V)$ and $\delta \log|y| = \mathrm{Re}(\epsilon/V)$, each with variance $\sigma^2/|V|^2$. Hence

$$\sigma_\psi^2 = \sum_{k\in\triangle} \frac{\sigma_k^2}{|V_k|^2}, \qquad \sigma_\ell^2 = \sum_{k\in\square} \frac{\sigma_k^2}{|V_k|^2}.$$

The first-order formulas are biased at low signal-to-noise, so visibilities enter closure phases only above SNR 2 and log closure amplitudes only above SNR 3.

6. Goodness of fit on closure data

Phases live on a circle. With a von Mises likelihood of concentration $\kappa = 1/\sigma_\psi^2$, the negative log-likelihood of an angular residual Δ is $\kappa(1-\cos\Delta)$ up to a constant, so we use

$$\chi^2_\psi = \frac{1}{N_\psi}\sum \frac{2\left(1-\cos(\psi^{\rm obs}-\psi^{\rm mod})\right)}{\sigma_\psi^2}, \qquad \chi^2_\ell = \frac{1}{N_\ell}\sum \frac{(\ell^{\rm obs}-\ell^{\rm mod})^2}{\sigma_\ell^2}.$$

Since $2(1-\cos\Delta) = \Delta^2 - \Delta^4/12 + \ldots$, the phase term equals the Gaussian Δ^2/σ^2 for small residuals and is periodic in Δ, so a model phase of 179° against data at -179° costs $(2^\circ)^2$, not $(358^\circ)^2$. The combined reduced statistic is $\chi^2_\nu = (N_\psi\chi^2_\psi + N_\ell\chi^2_\ell)/N$, $N = N_\psi + N_\ell$; "total" χ^2 means $N\chi^2_\nu$. In the code, $\cos(\psi^{\rm obs}-\psi^{\rm mod})$ is evaluated without any angle function: $\cos\Delta = (\mathrm{Re}\, z\cos\psi^{\rm obs} + \mathrm{Im}\, z \sin\psi^{\rm obs})/|z|$ for the model triple product z, which keeps gradients smooth.

7. The imaging assumptions

Every assumption is a differentiable map from parameters to an image of fixed flux, plus a regulariser R. With $p = I/F$ (unit-sum image) and a Gaussian prior image π of FWHM 40–45 μas, the regularisers are

$$R_{\rm MEM} = \sum_j p_j \log\frac{p_j}{\pi_j}, \qquad R_{\rm TSV} = N_{\rm pix}\sum\left[(\partial_x p)^2 + (\partial_y p)^2\right], \qquad R_{\rm TV} = \sum\sqrt{(\partial_x p)^2 + (\partial_y p)^2 + \varepsilon},$$

with forward differences. Imaging minimises $\chi^2_\nu(I) + R(I)$.

Assumption	Image as a function of parameters	Regulariser weights	Optimiser
RML-MEM	$I = F\,\mathrm{softmax}(\boldsymbol{\theta})$, one θ per pixel	MEM 0.5, TSV 0.1	L-BFGS
RML-TV	same	TV 0.6, MEM 0.2	L-BFGS
NF-1.5, NF-3.0	$I(x) \propto \exp\big(f_\theta(\gamma(x)) + \log \pi(x)\big)\, w(x)$	MEM 0.6 / 0.5, TSV 0.1 / 0.05	Adam
DIP	$I \propto \exp\big(G_\theta(z_0) + \log \pi\big)\, w$, fixed noise z_0	TSV 0.01	Adam
VAE-rings, VAE-knots	free pixels pulled to a trained generator $D(z)$	see 7.4	L-BFGS

Pixels are optimised with L-BFGS, as in eht-imaging: Adam normalises every coordinate's step, so pixels that the data barely constrain still move by about the learning rate per step and grow into hot spots. The weights were fixed once by a small parameter survey on synthetic M87*-like sources sampled exactly like the 2017 data (blurred normalised cross-correlation with the truth 0.78–0.89 for every assumption) and never changed afterwards.

7.1 Neural fields

Coordinates $x \in [-1, 1]^2$ are lifted by random Fourier features $\gamma(x) = F^{-1/2}[\cos 2\pi B x,\ \sin 2\pi B x]$, $B_{ij} \sim \mathcal{N}(0, s^2)$, $F = 48$ frequencies, and passed through a two-layer SiLU network of width 96. The bandwidth s (1.5 or 3.0 cycles per half field) sets the stationary kernel the network behaves like early in training, and thus how much fine structure it prefers. The soft window $w(x) = 1/(1 + e^{(r - 0.85\, r_{\max})/4\mu\mathrm{as}})$ removes emission at the edge of the field.

7.2 Deep image prior

A decoder with three stages (bilinear upsampling, two 3×3 convolutions with 16 channels and leaky ReLU) maps fixed noise on an 8×8 grid to 64×64. Only the weights are optimised. Batches of independent instances use grouped convolutions, so one call optimises several random starts at once.

7.3 Variational autoencoders trained on simulated black holes

Each training image is a thick ring $\exp\{-(r - r_0)^2 / 2s^2\}\,(1 + A\cos(\theta - \theta_b))$ with a faint interior floor, diameter 32–50 μas, width 6–15 μas, asymmetry 0–0.9, random bright position angle, a centring jitter of 2 μas and, for the "knots" model, with probability 1/2 one Gaussian knot of FWHM 6–12 μas carrying 2–15% of the flux at 26–52 μas from the centre. Each model sees 24,000 images. Encoder: four stride-2 convolutions (16, 32, 64, 96 channels) to a 24-dimensional latent; decoder: the mirror image with a sigmoid output. Training maximises the evidence lower bound with a Bernoulli likelihood and a KL weight β ramped to 0.5:

$$\mathcal{L}(\phi, \vartheta) = \mathbb{E}_{q_\phi(z|x)}\big[\log p_\vartheta(x \mid z)\big] - \beta\,\mathrm{KL}\big(q_\phi(z \mid x)\,\|\,\mathcal{N}(0, \mathrm{I})\big),$$

for 12 epochs with Adam and a cosine learning-rate schedule.

7.4 The learned prior as a regulariser

Searching only over the latent code z was too rigid: the best decoded images reached $\chi^2_\nu \approx 2.2$–2.9 on data whose true image has 0.90. Anelench therefore keeps a free pixel image and makes it pay for departing from what the generator can draw:

$$R_{\mathrm{VAE}}(I,z) = \lambda_f\, N_{\mathrm{pix}} \left\| p - \bar{D}(z) \right\|^2 + \lambda_z\, \tfrac{1}{2}\|z\|^2, \qquad \bar{D} = D/\textstyle\sum D, \quad \lambda_f = 1,\ \lambda_z = 0.02.$$

This is the negative log of a hierarchical prior: a Gaussian latent, and a Gaussian scatter of the image around the generator's output. With it the fits reach $\chi^2_\nu = 0.86$–0.99 and give the cleanest rings of all assumptions; the "rings" model visibly resists the knot because no training image had one.

8. The feature functional

A claim about an image is turned into a number at a common resolution. Let G be a circular Gaussian of FWHM $\phi = 20\,\mu\mathrm{as}$ ($s = \phi/2\sqrt{2\ln 2}$) and m a soft disk of radius r_R (edge width half a pixel) at the region centre. Then

$$q(I) = \frac{\langle G * I \rangle_m}{\max(G * I)}, \qquad \langle f \rangle_m = \frac{\sum_j m_j f_j}{\sum_j m_j}.$$

The claim under test is $q(I) > t$. The convolution is computed by FFT on a grid zero-padded to $2n \times 2n$: for output pixels and input pixels in $[0, n)$, their offset lies in $(-n, n)$, and the wrapped kernel on the $2n$ grid contains each such offset exactly once, so the circular convolution equals the linear one on $[0, n)^2$.

> **Proposition 6.** $q(cI) = q(I)$ for every $c > 0$, and $0 \le q \le 1$ for non-negative images.
>
> **Proof.** Convolution and the weighted mean are linear, so numerator and denominator both scale by c. The weighted mean of a non-negative function never exceeds its maximum. □

Scale invariance matters because closure quantities are blind to total flux; a feature measure that depended on it would be partly unmeasurable. Comparing at a common resolution prevents a sharp and a smooth explanation from "differing" only in sharpness.

9. When does an image "fit as well"?

9.1 The problem with the obvious rules

If the error bars are right, the true image has $\chi^2_{\mathrm{tot}} \sim \chi^2_N$: mean N, standard deviation $\sqrt{2N}$. Two obvious rules fail:

- **Goodness of fit**, $\chi^2_{\mathrm{tot}} \le N + 2\sqrt{2N}$: its slack grows with N (Proposition 8 below), so adding data that say nothing about the feature makes the acceptable set larger.
- **A fixed** $\Delta\chi^2$ above the best fit, e.g. 4: flexible image models also fit part of the noise, so the best fit lies *below* the true image's χ^2. On our synthetic data the truth sat 24–34 units above the best fits and would have been rejected.

9.2 Optimism

Consider the linear Gaussian picture in whitened units: data $\mathbf{y} = \boldsymbol{\mu} + \boldsymbol{\epsilon}$, $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathrm{I}_N)$, and a fit $\hat{\mathbf{y}} = H\mathbf{y}$ by a projection H (symmetric, idempotent, rank p) onto a model space containing $\boldsymbol{\mu}$.

Theorem 7. $\chi^2(\text{truth}) - \chi^2(\text{fit}) = \boldsymbol{\epsilon}^{\mathsf{T}} H \boldsymbol{\epsilon} \sim \chi^2_p$, independent of the residual $\chi^2(\text{fit}) \sim \chi^2_{N-p}$. In particular the truth sits on average p above the best fit, with standard deviation $\sqrt{2p}$.

Proof. $H\boldsymbol{\mu} = \boldsymbol{\mu}$, so $\mathbf{y} - \hat{\mathbf{y}} = (\mathrm{I} - H)\boldsymbol{\epsilon}$ and $\chi^2(\text{fit}) = \boldsymbol{\epsilon}^{\mathsf{T}}(\mathrm{I} - H)\boldsymbol{\epsilon}$, while $\chi^2(\text{truth}) = \boldsymbol{\epsilon}^{\mathsf{T}}\boldsymbol{\epsilon}$. The difference is $\boldsymbol{\epsilon}^{\mathsf{T}} H \boldsymbol{\epsilon}$. Diagonalising H with an orthogonal matrix turns these into sums of squares of disjoint sets of independent standard normals, p and $N - p$ of them (Cochran's theorem). □

For a general linear smoother the expected difference is $2\,\mathrm{tr}H - \mathrm{tr}H^{\mathsf{T}}H$ minus squared bias; we call it the optimism p_{eff} of the assumption. For regularised non-linear imaging it has no closed form, so we measure it.

9.3 Measuring optimism by parametric bootstrap

Treat the reference reconstruction $\hat{I}$ as if it were the truth, draw fresh closure noise around its predictions, $\psi^* = \psi(\hat{I}) + \mathcal{N}(0, \sigma^2_\psi)$, $\ell^* = \ell(\hat{I}) + \mathcal{N}(0, \sigma^2_\ell)$, refit from $\hat{I}$ to get $\hat{I}^*$, and set

$$\hat{p}_{\text{eff}} = \chi^2_{\text{tot}}(\hat{I};\, \mathbf{y}^*) - \chi^2_{\text{tot}}(\hat{I}^*;\, \mathbf{y}^*).$$

In the setting of Theorem 7 this is exactly a draw of χ^2_p, so it is unbiased for p. On synthetic data the estimates (26–31) agree with the true image's measured distance above the best fit (24–34).

9.4 The acceptance rule

Let k^* be the assumption with the lowest reference χ^2 and $p = \max(\hat{p}_{\text{eff}}(k^*), 2)$. An image is acceptable when

$$\chi^2_{\text{tot}}(I) \;\le\; \chi^2_{\text{tot,best}} + p + 2\sqrt{2p} \quad \text{and} \quad \chi^2_\nu(I) \le \max(\chi^2_{\nu,\text{best}}, 1)\left(1 + 2\sqrt{2/N}\right).$$

By Theorem 7 the true image passes the first condition with probability $P(\chi^2_p \le p + 2\sqrt{2p})$, about 0.97 for $p \approx 36$. The same ceiling applies to every assumption: how well an explanation must fit should not depend on how well its own optimiser happened to converge. After new data are added, p is not allowed to decrease (a model cannot absorb less noise from more data; a short refit can under-measure it).

Proposition 8 (why goodness of fit loosens). Let a wrong image have non-centrality λ, i.e. $\mathbb{E}\,\chi^2_{\text{tot}} = N + \lambda$. Under the goodness-of-fit rule its expected margin below the threshold is $2\sqrt{2N} - \lambda$. Adding n measurements that do not distinguish it from the truth leaves λ unchanged and raises the margin to $2\sqrt{2(N+n)} - \lambda$, so it is accepted more often. Under the optimism rule the margin is $p + 2\sqrt{2p} - \lambda$, which depends on the model's capacity and not on N.

10. The counterexample search

10.1 The question as two optimisation problems

For each assumption k with acceptable set $\mathcal{A}_k$,

$$q_k^- = \min_{I \in \mathcal{A}_k} q(I), \qquad q_k^+ = \max_{I \in \mathcal{A}_k} q(I).$$

The claim $q > t$ is **undetermined** if some assumption has an acceptable image with $q < t$ and some (possibly another) has one with $q > t$; **supported** if only the second kind exists; **rejected** if only the first.

10.2 Objective with penalty walls

Each instance with direction $s \in \{-1, +1\}$ minimises

$$\mathcal{J} = \chi^2_\nu + R - \beta\, r(t)\, s\, q + \frac{\kappa}{2}\left[\max\!\left(0,\ \frac{(\chi^2_\nu - c)\,N}{4}\right)^2 + \max\!\left(0,\ \frac{R - R_{\max}}{R_{\max}}\right)^2\right],$$

with push weight $\beta = 8$ ramped in by $r(t) \in [0, 1]$, wall stiffness $\kappa = 50$, the data wall c half a unit of total χ^2 inside the ceiling of 9.4, and $R_{\max} = 4R_{\mathrm{ref}}$, which only excludes wildly irregular images. Measuring the data excess in units of 4 (a 2σ change for one parameter) makes the wall equally stiff for any N. These one-sided quadratic walls replaced Lagrange multipliers updated by dual ascent, which exploded whenever an optimiser step overshot. Warm-started Adam instances use a learning rate reduced to 0.3 of nominal and ramped up over 60 steps, because a fresh Adam moves every weight by about the learning rate in its first steps and knocks a converged network off its fit.

10.3 Starting points

- **Twins.** From each of an assumption's two best reference reconstructions, one instance pushes q down and one pushes it up. The two results share their starting image, so they differ mainly where the claim is.
- **Seeded starts** (pixel assumptions). The reference with the region multiplied by $1 - 0.95\,m'$ or $1 + 1.5\,m'$, where m' is a soft disk 3 μas larger than the region, projected back onto the data by the walls. A local push from a reconstruction can miss explanations in another basin.
- **Carried witnesses** (after new data). The most extreme acceptable explanation per assumption and direction from the previous search, which must now also fit the new night.

Every iterate is checked against both constraints; each instance keeps its most extreme acceptable iterate.

> **Proposition 9 (what the reported range means).** Let $\hat{q}^-$, $\hat{q}^+$ be the extremes over all acceptable explanations found. Then $[\hat{q}^-, \hat{q}^+] \subseteq [\min_k q_k^-,\ \max_k q_k^+]$.
>
> **Proof.** Every reported explanation satisfies the acceptance constraints of its assumption, so its q lies in that assumption's attainable range. □
>
> **Corollary.** "Undetermined" is certified: two explicit acceptable images on opposite sides of t are exhibited. "Supported" and "rejected" mean that no counterexample was found under the explored assumptions and starts, not that none exists.

11. Observation design

11.1 Feasible nights

A package a is a set of stations and a UT window. Scans of 6 minutes are placed every 12 minutes; a baseline is measured in a scan when both stations see M87* above 10° elevation; intra-site baselines are

excluded; both 2 GHz bands are used. Its thermal noise follows 5.1 with the effective SEFDs of 5.2, plus the 2% systematic term evaluated on the reference image's amplitudes. Its cost is the window length times the sum of station weights (ALMA 3, NOEMA 2, SMA 1.5, others 1). The candidate set has ten station sets (the 2017 array, with GLT, Kitt Peak and NOEMA in all combinations, and three sub-arrays) × 3 h and 6 h windows × twelve start times; for real data the three held-back 2017 nights are added with their real sampling.

11.2 Expected separation between two explanations

For a package, let $\psi^{(0)}, \ell^{(0)}$ and $\psi^{(1)}, \ell^{(1)}$ be the closure quantities that explanations I_0 and I_1 predict, and

$$d_a(I_0, I_1)^2 = \sum \frac{2\left(1 - \cos(\psi^{(0)} - \psi^{(1)})\right)}{\sigma_\psi^2} + \sum \frac{(\ell^{(0)} - \ell^{(1)})^2}{\sigma_\ell^2}.$$

Proposition 10. In the Gaussian approximation, if the new data are generated by I_1, the statistic $\Delta = \chi^2_{\mathrm{tot}}(I_0) - \chi^2_{\mathrm{tot}}(I_1)$ on those data is distributed as $\mathcal{N}(d^2,\, 4d^2)$. Hence d is the expected separation in units of the noise, and the probability that the new night favours the right explanation by more than Δ_0 is $\Phi\big((d^2 - \Delta_0)/2d\big)$.

Proof. With whitened data $\mathbf{y} = \mathbf{F}_1 + \boldsymbol{\epsilon}$: $\Delta = \|\mathbf{F}_1 - \mathbf{F}_0 + \boldsymbol{\epsilon}\|^2 - \|\boldsymbol{\epsilon}\|^2 = \|\mathbf{F}_1 - \mathbf{F}_0\|^2 + 2(\mathbf{F}_1 - \mathbf{F}_0)^{\mathsf{T}}\boldsymbol{\epsilon}$, a constant d^2 plus a Gaussian of variance $4d^2$. □

For $\Delta_0 = 4$: $d = 3$ gives 0.80, $d = 5$ gives 0.98.

11.3 The criterion

With witness pairs $\mathcal{W}$ (twins when available, otherwise all cross pairs) and a budget B,

$$a^* = \arg\max_{a:\, c(a) \le B,\, d_{\min}(a) \ge 3} \frac{d_{\min}(a)}{c(a)}, \qquad d_{\min}(a) = \min_{(I_0, I_1) \in \mathcal{W}} d_a(I_0, I_1),$$

falling back to the largest $d_{\min}$ when no package reaches 3σ. The maximin over pairs asks a night to separate *every* pair, including the hardest. A variant spends the whole budget: $\arg\max_{c(a) \le B} d_{\min}(a)$.

The criterion is computed over a finite pool of fixed explanations. After the night, explanations can adapt to the new data, which the pool does not anticipate; this is why every recommendation is checked by adding the night and running the whole search again.

11.4 Comparison strategies

Strategy	Chooses the feasible night that maximises
Random	nothing: uniform over nights within the budget
Highest signal-to-noise	$\sum_k \lvert V_k(\hat{I})\rvert/\sigma_k$ over the night's visibilities
Largest new Fourier coverage	$\sum_k \min\big(1,\ \mathrm{dist}(\mathbf{u}_k,\ \mathrm{existing} \pm \mathbf{u})/\delta u\big)$, with $\delta u = 1/\mathrm{FOV}$
Single-model uncertainty	mean pairwise separation d_a among one assumption's restarts
Anelench	$d_{\min}(a)/c(a)$ as above

12. Algorithms

Algorithm 1. Closure basis for one scan and band

1. Collect baselines with data; mark those above SNR 2 (phases) and SNR 3 (amplitudes).
2. Form every triangle with three marked baselines; its vector has $+1, +1, -1$ on ab, bc, ac; its noise is σ_ψ.
3. Form every quadrangle and every pair of its three matchings with all four baselines marked; vector $+1$ on the numerator, -1 on the denominator; noise σ_ℓ.
4. For each kind separately: sort by noise; keep a candidate if its Gram–Schmidt residual against the kept vectors exceeds 10^{-8}.

Algorithm 2. Reference imaging under one assumption

1. Create a batch of independent instances (two restarts; a third warm-started from the previous reconstruction after new data).
2. Minimise $\chi^2_\nu + R$ for 900 steps (Adam) or 90 outer × 10 inner L-BFGS iterations with strong Wolfe line search.
3. Sort the instances by χ^2; keep their parameters for warm starts.

Algorithm 3. Optimism by bootstrap

1. Predict closure phases and log closure amplitudes of the best reference $\hat{I}$.
2. Add fresh Gaussian noise with the measured σ_ψ, σ_ℓ; refit for 400 steps from $\hat{I}$.
3. Return $\max\{0,\ \chi^2_{\mathrm{tot}}(\hat{I}) - \chi^2_{\mathrm{tot}}(\hat{I}^*)\}$ on the bootstrap data.

Algorithm 4. Counterexample search

1. Compute the ceiling of 9.4 from the best reference fit and its optimism.
2. For each assumption, build instances: twins from the two best references; seeded starts for pixel assumptions; carried witnesses if any.
3. Minimise $\mathcal{J}$ of 10.2 for 600 steps; after each evaluation, if the iterate is acceptable and more extreme in its direction than the kept one, keep it.
4. Report every kept explanation, the range of q, the verdict of 10.1 and the twin pairs.

Algorithm 5. Planning and re-testing

1. For every candidate night: compute its sampling and noise; select its closure basis using the reference image's predicted amplitudes; predict every witness's closures; compute $d_{\min}$, signal-to-noise, coverage gain and single-model spread.
2. Choose the night by the criterion of 11.3 (and by each comparison strategy).
3. Re-test: simulate the night from the hidden truth with new station gains, the 2% non-closing error and thermal noise (synthetic data), or add the real night (real data); concatenate with the old data; re-image with warm starts; rerun the search with carried witnesses.

Costs. A forward evaluation is two dense real matrix products of size $n_{\text{vis}} \times N_{\text{pix}}$ (a few hundred by 4096), cheap enough that every assumption is optimised as a batch on a laptop CPU: with seven assumptions the reference images take one to four minutes, a search two to four minutes, and the scoring of 173 nights under a minute.

13. Verification

- Closure phases and log closure amplitudes are unchanged, to 10^{-9}, when random complex gains multiply every visibility (Theorems 1, 2).
- For complete arrays of 4–7 stations the selected bases have exactly $(S-1)(S-2)/2$ phases and $S(S-3)/2$ amplitudes (Theorems 3, 4).
- (u, v) recomputed from station positions, GMST and the hour angle reproduce the release on every baseline to a relative error below 10^{-3} (measured 3×10^{-5}).
- The forward model reproduces noiseless closures of a known image to far below the noise, and a different image is rejected.
- The feature value is invariant to image scaling and increases when a knot is added (Proposition 6).
- Every image model can export one instance's state and import it into another without touching the rest of the batch; this is what warm starts and carried witnesses rely on.

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